

case study 2\main.py

```
1 patient_name = input("patient name")
2 department_name = input("Department name
:cardiology,neurology,orthopedics,dermatology,gynecology").split(",")
3 available_departments = input("available departments :
cardiology,neurology,orthopedics,dermatology,gynecology").split(",")
4 previously_visited_departments = input("previously visited departments :
cardiology,neurology,orthopedics,dermatology,gynecology").split(",")
5 preferred_doctor = input("preferred doctor : Dr. Kumar, Dr. Sharma, Dr. Khan , Dr. Smith, Dr.
Donald").split(",")
6 available_doctors = input("available doctors : Dr. Kumar, Dr. Sharma, Dr. Khan , Dr. Smith,
Dr. Donald").split(",")
7 emergency_departments = input("emergency_departments:Cardiology , Orthopedics ").split(",")
8
9 requested_department_set = set(department_name)
10 available_department_set = set(available_departments)
11 visited_set = set(previously_visited_departments)
12 preferred_doctor_set = set(preferred_doctor)
13 available_doctors_set = set(available_doctors)
14 emergency_departments_set = set(emergency_departments)
15
16
17 common_departments = requested_department_set & available_department_set
18 common_doctors = preferred_doctor_set & available_doctors_set
19
20 unavailable_departments = requested_department_set - available_department_set
21 unavailable_doctors = preferred_doctor_set - available_doctors_set
22
23
24 previously_visited_departments = requested_department_set & visited_set
25
26
27 emergency_departments_requested = requested_department_set & emergency_departments_set
28
29
30 if len(department_name) != len(requested_department_set):
31     duplicate_request = True
32
33 else:
34     duplicate_request = False
35
36 if len(emergency_departments_requested) > 0:
37     alloted_departments = list(emergency_departments_requested)[0]
38
39 elif len(common_departments) > 0:
40     alloted_departments = list(common_departments)[0]
41
42 else:
43     alloted_departments = "No department available"
```

```
44
45
46 if len(emergency_departments_requested) > 0:
47     appointment_status = "Emergency appointment"
48 elif len(common_departments) > 0:
49     appointment_status = "Appointment confirmed"
50 else:
51     appointment_status = "Appointment not available"
52
53
54 print("Patient Name:", patient_name)
55 print("Requested Departments:", department_name)
56 print("Available Departments:", available_departments)
57 print("Previously Visited Departments:", previously_visited_departments)
58 print("Preferred Doctors:", preferred_doctor)
59 print("Available Doctors:", available_doctors)
60 print("Emergency Departments:", emergency_departments)
61
62 print("Common Departments:", common_departments)
63 print("Common Doctors:", common_doctors)
64
65 print("Unavailable Departments:", unavailable_departments)
66 print("Unavailable Doctors:", unavailable_doctors)
67
68 print("Previous Visited Departments:", previously_visited_departments)
69
70 print("Duplicate Request:", duplicate_request)
71 print("Alloted Departments:", alloted_departments)
72 print("Appointment Status:", appointment_status)
```